

Credit Risk Modeling and Expected Loss Prediction Using Logistic Regression

Introduction

Credit risk is one of the most important risks faced by financial institutions. Before approving a loan, banks must assess the likelihood that a borrower may fail to repay the loan. This likelihood is known as the Probability of Default (PD).

The objective of this project is to build a machine learning model that predicts the probability of default for a borrower using historical loan data. The predicted probability is then used to estimate the expected loss associated with a loan.

Objective

The main objectives of this project are:

- Predict the Probability of Default (PD) for loan borrowers.
 - Build a classification model using historical loan data.
 - Calculate Expected Loss (EL) for a loan.
 - Assist financial institutions in making informed lending decisions.
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Dataset Description

The dataset contains borrower information and loan-related attributes.

Features Used

- Credit Lines Outstanding
- Loan Amount Outstanding
- Total Debt Outstanding
- Income
- Years Employed
- FICO Score

Target Variable

- Default
- 0 = No Default
- 1 = Default

The target variable indicates whether a borrower has previously defaulted on a loan.

Methodology

The project follows a supervised machine learning approach.

Step 1: Data Loading

The dataset was loaded into Python using the Pandas library.

Step 2: Data Exploration

The following checks were performed:

- Dataset structure using `df.info()`
- Statistical summary using `df.describe()`
- Missing value analysis using `df.isnull()`

Step 3: Feature Selection

The customer identifier column was removed because it does not contribute to predicting default risk.

The remaining borrower attributes were used as input features.

Step 4: Train-Test Split

The dataset was divided into:

- Training Data (80%)
- Testing Data (20%)

This allows the model to learn from one portion of the data and be evaluated on unseen records.

Model Selection

Logistic Regression was selected for this project because:

- The target variable is binary.
- It predicts probabilities directly.
- It is widely used in credit risk modeling.
- It provides interpretable results.

The model was trained using the training dataset and evaluated using the testing dataset.

Probability of Default (PD)

The trained model predicts the probability that a borrower will default.

Example:

If the model predicts:

$$PD = 0.25$$

This means the borrower has a 25% probability of defaulting on the loan.

Expected Loss Calculation

The project assumes a Recovery Rate of 10%.

Recovery Rate represents the percentage of money recovered after a borrower defaults.

Loss Given Default (LGD)

$$LGD = 1 - \text{Recovery Rate}$$

$$LGD = 1 - 0.10$$

$$LGD = 0.90$$

Expected Loss Formula

$$\text{Expected Loss} = PD \times \text{Loan Amount} \times LGD$$

Where:

- PD = Probability of Default
 - Loan Amount = Outstanding Loan Amount
 - LGD = Loss Given Default
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Results

The Logistic Regression model successfully predicted the probability of default for borrowers based on their financial characteristics.

The model can estimate:

- Borrower default risk
- Expected financial loss
- Risk level associated with each loan application

These predictions can support lending decisions and risk management strategies.

Business Impact

This model can help financial institutions:

- Reduce credit risk exposure
 - Improve loan approval decisions
 - Identify high-risk borrowers
 - Estimate potential financial losses
 - Strengthen portfolio risk management
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Conclusion

This project successfully developed a credit risk prediction model using Logistic Regression. The model estimates the Probability of Default (PD) for borrowers and calculates the Expected Loss (EL) associated with a loan.

By combining machine learning techniques with financial risk metrics, the model provides a practical framework for assessing borrower risk and supporting data-driven lending decisions.